

PLASTIC-CRAFT

Noryl

Modified Polyphenylene Oxide (PPO / PPE + PS)

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One of the most dimensionally stable thermoplastics available. Outstanding electrical insulation over a wide range of humidity and temperature. Inherently flame retardant, hydrolytically stable, and electroplatable — an ideal lower-cost alternative to Ultem® in many electrical applications.

STOCK AVAILABILITY

Plastic-Craft maintains extensive inventory across a wide range of standard and specialty sizes in sheets, rods, tubes, and custom profiles. Most orders ship from stock — contact us for immediate availability.

KEY HIGHLIGHTS

Superior dimensional stability
Excellent electrical insulation
Inherently flame retardant (V-1)
Very low moisture absorption
Low creep & high stiffness
Electroplatable

✓ Stable in boiling water — ideal for steam & autoclave

COMMON APPLICATIONS

Electrical housings & insulators
Pump & valve components
Medical device housings
Business machine parts
Fluid handling systems
Automotive dashboards
Appliance internals

GRADES AVAILABLE

Standard (black)
SE1 (V-0 flame retardant)
GFN series (glass-filled)
EN-265 (extruded shapes)
Non-halogen FR grades

MATERIAL PROPERTIES — TYPICAL VALUES

PROPERTY	ASTM	TYPICAL VALUE	UNITS
PHYSICAL			
Specific Gravity	D792	1.06	—
Water Absorption, 24h	D570	0.06	%
MECHANICAL			
Tensile Strength, Ultimate	D638	9,600	psi (66 MPa)
Tensile Modulus	D638	350,000	psi (2,410 MPa)
Elongation at Break	D638	28	%
Flexural Strength	D790	14,400	psi (99 MPa)
Flexural Modulus	D790	370,000	psi (2,550 MPa)
Compressive Strength	D695	13,000	psi (90 MPa)
Impact, Izod Notched	D256	5.0	ft-lb/in (267 J/m)
Hardness, Rockwell	D785	R115	—
THERMAL			
Heat Deflection @ 264 psi	D648	265	°F (129°C)
Heat Deflection @ 66 psi	D648	285	°F (141°C)
Max Continuous Service	—	220	°F (104°C)
CTE (Linear)	D696	3.3 × 10 ⁻⁵	in/in/°F
Thermal Conductivity	C177	1.5	BTU·in/hr·ft²·°F
Glass Transition (Tg)	—	~290	°F (143°C)
ELECTRICAL			
Volume Resistivity	D257	> 10 ¹⁷	ohm·cm
Surface Resistivity	D257	> 10 ¹⁶	ohm/sq
Dielectric Strength	D149	500	V/mil (20 kV/mm)
Dielectric Constant, 1MHz	D150	2.7	—
Dissipation Factor, 1MHz	D150	0.0008	—
FLAMMABILITY			
UL Flammability Rating	UL 94	V-1	V-0 grades available
DURABILITY			
Abrasion Resistance (Taber)	D1044	10	mg/1000 cycles
Weathering / UV	G154	Fair	UV-stab. grades avail.

PROCESSING / FABRICATION NOTES: Excellent machinability — machines to tight tolerances | One of few thermoplastics that can be electroplated | Bondable with solvent cements & adhesives | Weldable by ultrasonic, hot plate & vibration methods

CHEMICAL RESISTANCE: Resistant to water, salt solutions, dilute acids & bases. Hydrolytically stable. **AVOID:** aromatic & chlorinated hydrocarbons, gasoline, kerosene, methylene chloride — may cause stress cracking.

Typical values; not to be used as specifications. Actual values may vary by grade, manufacturer, and processing conditions. Contact Plastic-Craft for grade-specific data.